Quantum Mechanics I — Exercise Sheet 14

Identical particles

Variant: Questions only

Conventions, as in Lecture 14: the swap $W(u \otimes w) = w \otimes u$; $\mathrm{Sym}^2 V$ and $\Lambda^2 V$ its eigenspaces for $+1$ and -1; at a beam splitter, the ports e_1, e_2, one unitary U acting on each particle, and the coincidence projection $E_c = E_1 \otimes E_2 + E_2 \otimes E_1$ with $E_i = |e_i\rangle\langle e_i|$.

Exercise 1 (the two sectors). Let $u_1, \dots, u_n$ be an orthonormal basis of V. Show that the vectors $u_i \otimes u_i$ and $\frac{1}{\sqrt{2}}(u_i \otimes u_j + u_j \otimes u_i)$, $i < j$, form an orthonormal basis of $\mathrm{Sym}^2 V$, and the vectors $\frac{1}{\sqrt{2}}(u_i \otimes u_j - u_j \otimes u_i)$, $i < j$, of $\Lambda^2 V$. Conclude that $\dim \mathrm{Sym}^2 V = \frac{1}{2}n(n+1)$ and $\dim \Lambda^2 V = \frac{1}{2}n(n-1)$. Finally, for $V = \mathbb{C}^4$ with the standard basis $e_0, \dots, e_3$, list the basis of $\Lambda^3 \mathbb{C}^4$ given by Proposition 14.7, and write $e_0 \wedge e_1 \wedge e_2$ as a sum of products.

Exercise 2 (the Hong–Ou–Mandel dip). Two photons enter the two ports of a balanced splitter. Their internal states, polarisation and wave packet, are the unit vectors α and β of an internal space $\mathbb{C}^d$, with the overlap $s = \langle \alpha, \beta \rangle$; the splitter acts as $U \otimes \mathbb{1}$ on each photon. Show that the coincidence probability is $\frac{1}{2}(1 - |s|^2)$. What would two fermions give?

Exercise 3 (two bosons in one port). Two bosons enter the same input port, in the state $e_1 \otimes e_1$, of a balanced splitter. Find the coincidence probability. Can two fermions be prepared in this way?

Exercise 4 (two electrons on a ring). Two electrons hop on the ring of four sites of Lecture 8, with the one-electron Hamiltonian $h = c\mathbb{1} - \kappa(T + T^\dagger)$ acting on the orbital part and the identity on the spin. They do not interact with each other, so the pair has $H = h \otimes \mathbb{1} + \mathbb{1} \otimes h$ on $\Lambda^2(\mathbb{C}^4 \otimes \mathbb{C}^2)$. Find the lowest energy and its states, and the next energy and its degeneracy.

Exercise 5 (two electrons on two sites: the limits). Take the model (14.2), with $\kappa > 0$, and $\Delta = \sqrt{\varepsilon_U^2/4 + 4\kappa^2} - \varepsilon_U/2$, the splitting of Proposition 14.8.

(a) For $\varepsilon_U = 0$, show that the lowest state is the Slater determinant $(f_+ \otimes |\uparrow\rangle) \wedge (f_+ \otimes |\downarrow\rangle)$ of the bonding orbital $f_+ = \frac{1}{\sqrt{2}}(e_a + e_b)$, with the energy -2κ, and find its ionic weight (the probability that both electrons are on one site).

(b) For $\varepsilon_U > 0$, find the lowest eigenvector of the matrix of Proposition 14.8 and its ionic weight. What happens as $\varepsilon_U \to \infty$?

(c) Expand Δ in powers of κ/ε_U to the second nonzero term.

(d) Show that the four lowest levels are those of the spin Hamiltonian $\Delta\big(\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)}/\hbar^2 - \frac{1}{4}\big)$, for two spins one half attached to the sites. Is this an identity between H and that matrix?

(e) Show that for $\varepsilon_U > 0$ the lowest state is not a single Slater determinant $u \wedge w$, $u, w \in V$.

Exercise 6 (⋆ an unbalanced splitter). A lossless splitter transmits with probability T and reflects with probability $R = 1 - T$: $U = \left(\begin{smallmatrix} \sqrt{T} & \mathrm{i}\sqrt{R} \\ \mathrm{i}\sqrt{R} & \sqrt{T} \end{smallmatrix}\right)$. Find the coincidence probabilities for two bosons, two fermions and two distinguishable particles, one at each input. How deep is the bosons' dip, $(P_{\mathrm{dist}} - P_{\mathrm{bos}})/P_{\mathrm{dist}}$?